

UPI Transaction Analytics

DATA ANALYTICS WITH GENAI · CAPSTONE PROJECT

A Simple Report on What We Did and What We Found

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1. What this project is about?

Many people in India use UPI to send and receive money every day. With so many payments happening, companies need to know two things: which payments fail, and where fraud happens the most.

In this project, we took a large set of UPI transaction data and worked with it step by step. First we cleaned it. Then we stored it properly. Then we studied it using Python. Then we tested our ideas with statistics. Finally, we built simple dashboards so anyone can look at the results, even people who don't know coding.

The goal was simple: turn messy data into clear answers that a business team can use to stop fraud and make payments smoother.

2. The Data we used

We used 7 tables of data. These tables held information about customers, their devices, their UPI accounts, merchants, transactions, customer feedback, and fraud alerts. Together, this added up to 140,500 records.

Table Name	Number of Records
Customer_Master	10,000
Device_Info	12,000
UPI_Account_Details	12,000
Merchant_Info	500
UPI_Transaction_History	100,000
Customer_Feedback_Surveys	4,000
Fraud_Alert_History	2,000
Total Records	140,500

Before starting any analysis, we checked that these tables matched each other correctly. For example, every transaction had to belong to a real customer and a real device.

How we organized the Data

Instead of keeping the data in separate, messy files, we built one connected database. Every table links to the others using ID numbers. For example, each transaction is linked to a customer id or a device id.

This makes it easy to ask questions like: “Which devices, in which regions, had the most fraud?” We also added simple rules to keep the data clean. For example: a customer’s age must be 18 or older, and a risk score must be between 0 and 1. If new data breaks these rules, it gets rejected automatically.

Tools we used

- **Excel** — to look at the raw data and spot early mistakes.
- **SQL (MySQL)** — to store all 140,500 records safely in one connected database.
- **Python** — to clean the data further, explore it, and run statistical tests.
- **Power BI** — to turn the results into easy dashboards anyone can use.

3. Cleaning the Data

Raw data almost always has mistakes. Before we could trust it, we had to clean it up. Here is what we did:

- **Fixed spelling mistakes** — for example, “succes” was changed to “Success”, and “Androd” was changed to “Android”.
- **Flagged bad data instead of deleting it** — 9.69% of customers had a mobile number with the wrong number of digits. We marked these for review instead of removing them, so no data was lost.
- **Filled in missing values** — for example, if a transaction had no merchant, we labeled it as a person-to-person transfer instead of leaving it blank.
- **Checked for duplicate records** — we found zero duplicates across all 7 tables. This told us the data was reliable.

After cleaning, we loaded the data into SQL with safety rules built in, so bad data could not enter the system again later.

4. What we Find?

Once the data was clean and ready in Python, a few clear patterns showed up:

- **Failed payments:** 5.87% of all transactions failed. That's a big number, because every failed payment can make a customer lose trust.
- **Fraud is rare, but not spread evenly:** overall, about 2% of transactions were fraud. But the West region had the highest fraud rate (2.09%), and Feature Phones had the highest fraud rate by device (2.15%).
- **Some shopping categories are riskier:** Electronics (2.24%) and Apparel (2.22%) had the most fraud. This is likely because these purchases cost more money, which attracts scammers.
- **Usage is growing fast:** the number of transactions grew about 18 times during the period we studied. Fraud grew much slower than that, which is a good sign — but it still needs to be watched closely.

5. Testing our ideas with Statistics

We didn't want to just guess why fraud happens. So we ran a few statistical tests to check if things like device type or merchant category were closely linked to fraud. Here is what we found, in simple words:

Test	What We Checked	p-value	Result
T-Test	Do fraud and non-fraud transactions have different average amounts?	0.933	No difference found
ANOVA	Does average transaction amount change across merchant types?	0.454	No difference found
Chi-Square	Is fraud more common on certain device types?	0.129	No link found
Correlation	Does a customer's risk score relate to fraud happening?	0.612	No link found

In simple words: none of these tests found a strong, clear link between fraud and any single factor by itself. This does not mean our earlier findings are wrong. It means fraud is probably caused by a mix of things happening together — like one device, in one region, buying one type of product — rather than just one factor alone. This is still a useful result. It tells the

business that simple rules, like “block all Feature Phone users,” will not work well. A smarter approach that looks at many factors together is needed instead.

6. Dashboard we Built

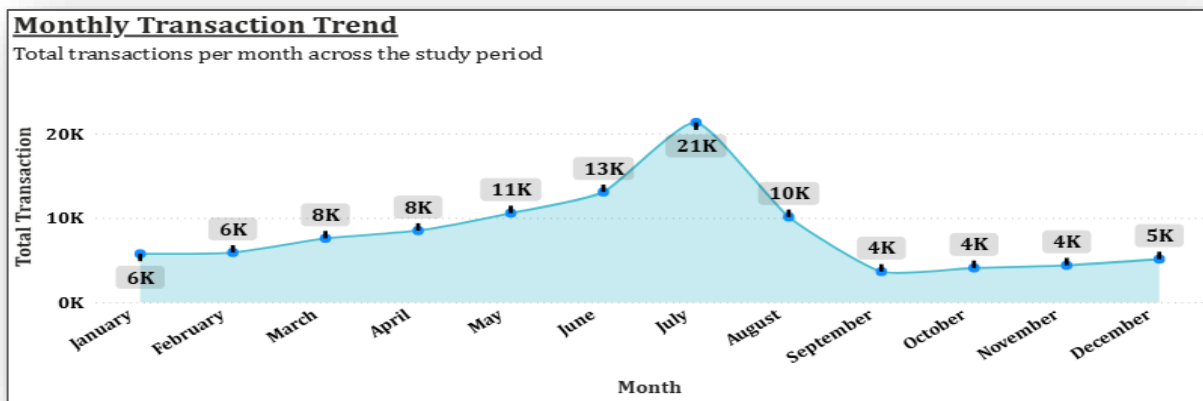
To make these findings easy to use, we built three Power BI dashboard pages:

- **Executive Dashboard** — shows total transactions, average amount, failure rate, fraud rate, and trends over time. It can be filtered by region, device, and merchant type.
- **Fraud & Operations Dashboard** — shows fraud alert trends, the causes behind alerts, how long alerts take to resolve, and which devices or merchants are riskiest.
- **Alert Details** — lets you click into any single fraud alert to see the exact customer, merchant, transaction, and how it was resolved.

7. A Closer look at the dashboard's Key charts

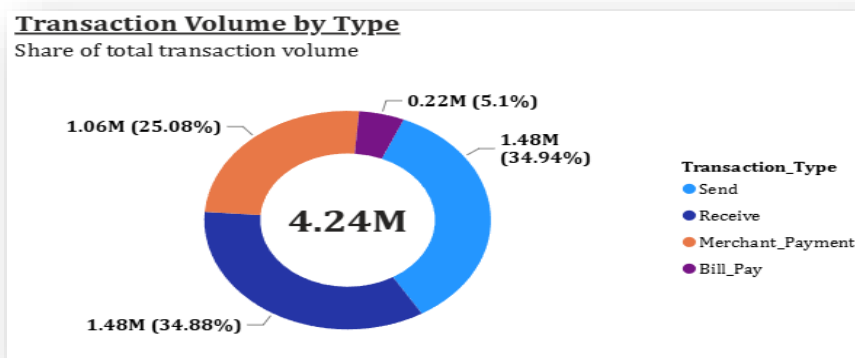
The dashboard has a lot on it, so here is a plain-language walkthrough of the charts that matter most and what each one is actually telling us.

Monthly Transaction Trend



This line chart shows how many transactions happened each month, from about 6,000 in January up to a sharp peak of 21,000 in July, before dropping back down to a steady 4,000–5,000 a month. A spike this sharp is worth double-checking — it could be a real seasonal event (like a festival or sale period), or it could be a one-off data artifact. Either way, it's the first thing we'd confirm with the business team before making capacity or staffing decisions around it.

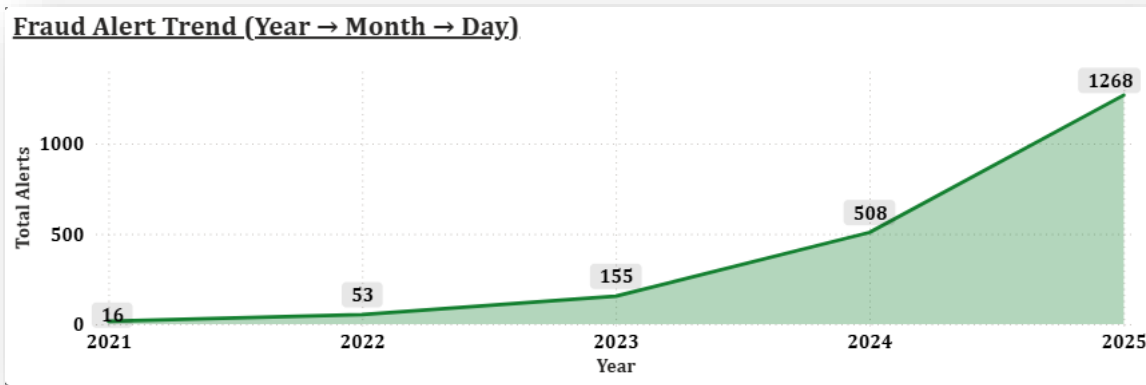
Transaction Volume by Type



This donut chart splits all transactions into four types: Send (34.94%), Receive (34.88%), Merchant Payment (25.08%), and Bill Pay (5.1%). In plain terms, roughly 70% of all activity on the platform is people sending or receiving money from each

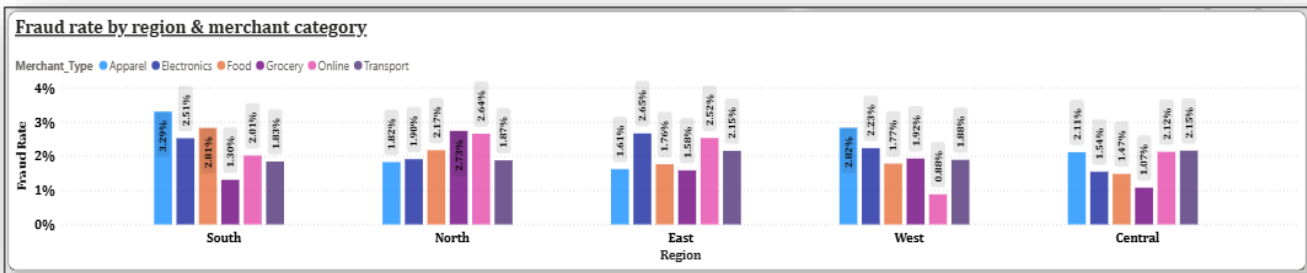
other — not shopping. Only about a quarter of activity is actual merchant payments. This matters because it tells us where the fraud risk we found earlier (in Electronics and Apparel purchases) is concentrated: inside that smaller 25% slice, not spread across the whole platform.

Fraud Alert Trend by Year



This chart shows how many fraud alerts were raised each year: just 16 in 2021, climbing to 53, then 155, then 508, and finally 1,268 in 2025. That’s roughly an 80-times increase over five years — a much steeper climb than regular transaction growth. This tells us fraud monitoring needs to grow ahead of the platform, not just alongside it, or the fraud team will keep falling further behind.

Fraud Rate by Region and Merchant Category



This chart breaks fraud rate down by region and by what kind of merchant the money went to. The standout finding here is the West region combined with Electronics merchants, which reaches a 3.01% fraud rate — close to double the platform's overall 2% average. No other region-and-category combination comes close. This is the single most useful chart for deciding where to focus fraud-prevention resources first.

Device Fraud Rate — Rooted vs Non-Rooted

This table compares fraud rates on rooted versus non-rooted devices, for every device type. The pattern is consistent and striking: rooted devices show a fraud rate 15–20 times higher than non-rooted ones, across every single device type.

Device Type	Rooted	Fraud Rate	Fraud Txns	Total Txns
Android	False	0.01%	345	24,422
Android	True	0.22%	156	723
Feature Phone	False	0.01%	359	24,635
Feature Phone	True	0.20%	192	961

iOS	False	0.01%	319	24,087
iOS	True	0.20%	141	696
Tablet	False	0.01%	321	23,685
Tablet	True	0.21%	167	791

For example, a non-rooted Feature Phone has a fraud rate of just 0.01%, but a rooted Feature Phone jumps to 0.20%. The same jump happens for Android, iOS, and Tablet. This is one of the clearest, most actionable patterns in the entire dataset — rooted-device detection alone could meaningfully cut fraud exposure.

High-Risk Merchants

This table lists the merchants with the highest risk scores — a combined measure the platform uses to flag merchants worth reviewing. A sample of the highest-risk merchants currently flagged:

Merchant Name	Category	Region	Risk Score
Armstrong-Brown	Apparel	East	0.50
May-Schultz	Apparel	East	0.50
Stevens-Cole	Food	South	0.50
Ho Plc	Electronics	South	0.49
Brown Inc	Food	Central	0.38
Brown Inc	Online	West	0.24

Most of the merchants on this list fall into Apparel, Food, or Electronics — the same categories flagged earlier — which adds more confidence to the recommendation to review on boarding and verification for these merchant types specifically.

Alert Details — Drill-Through Record View

This is a detail page, not a summary chart — it lets the fraud team click into any single alert and see exactly what happened: the customer, the merchant (if any), the transaction amount, and whether and when it was resolved. Many of the rows show “No Merchant,” meaning the flagged activity was a person-to-person transfer, not a purchase — useful context, since it means not all fraud alerts are about shopping fraud; some are about suspicious login or transfer timing instead.

8. What we Recommend

- **Add extra security for Feature Phone users**, such as a phone call to confirm big transactions. This device type has the highest fraud rate.
- **Check merchants more carefully in Electronics and Apparel**, especially in the West region, since fraud is highest there (3.01% in that combination alone).
- **Treat rooted devices as a distinct risk group**, since fraud rates are 15–20 times higher on rooted devices across every device type — this is one of the strongest patterns in the data.
- **Fix the mobile number problem**, by asking the 9.69% of customers with wrong numbers to update their details.

- **Stop relying on single-factor rules**, and instead build a risk score that looks at region, device, and merchant type together.
- **Run a customer education campaign** to bring down the 5.87% failure rate, since many failures are likely due to simple mistakes like a wrong PIN.

9. A Few things to keep in mind

No study is perfect. Here are a few limits to remember:

- **Fraud cases are rare** — only about 2% of transactions were fraud. When one group is this small, it's harder for tests to find patterns, even real ones.
- **We ran several tests together** — when you run many tests at once, there's a small chance one looks meaningful just by luck. We did not correct for this.
- **This is based on one time period** — fraud patterns can change over time, so these findings should be checked again regularly.

10. Conclusion

This project shows how messy, raw data can be turned into something useful, step by step. We cleaned the data, stored it properly, tested our ideas honestly, and built simple dashboards. Together, this gives the UPI team a solid starting point to reduce fraud and improve the customer experience.

The biggest lesson: fraud is not caused by just one thing. It shows up more when certain things happen together — a certain device, in a certain region, buying a certain type of product. Any future plan to stop fraud should keep this in mind.